

Kristopher Ali

Senior Data Engineer

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Profile

- **Senior Data Engineer** with 10+ years of experience building large-scale data platforms, pipelines, and analytics systems across healthcare, telecom, and government domains.
- Expert in **modern cloud data stacks** – including Azure, Snowflake, Databricks, GCP, dbt, and Apache Spark – with a strong track record of enabling ML/AI workflows and real-time insights.
- Deep domain expertise in **healthcare data engineering**, working with clinical, claims, EHR, and patient engagement data under strict compliance (HIPAA, HEDIS, CMS, SOC2).
- Proven leader in **platform modernization**, **data quality governance**, and **cross-functional collaboration**, mentoring junior engineers and acting as SME for clients like Mayo Clinic, HCA Healthcare, and Comcast.
- Passionate about turning complex data into clean, reusable, and analytics-ready assets that power better decisions, predictive models, and improved outcomes.

Education

University of Michigan, Bachelor of Arts(B.A.), Informatics

Aug 2009 – May 2013 | Ann Arbor, MI

Professional Experience

Deloitte, Senior Data Engineer

Mar 2022 – Mar 2026 | Remote

Worked with two major U.S. healthcare provider clients – **Mayo Clinic** and **HCA Healthcare** – to modernize clinical data platforms, enable real-time analytics and ML pipelines, and accelerate data-driven care delivery across their hospital systems. Focused on building robust, HIPAA-compliant, cloud-native infrastructure using **Azure, Databricks, and Snowflake**, supporting high-scale patient care, operational analytics, and AI-readiness across inpatient and outpatient systems.

• Data Platform Modernization & Architecture

- Migrated 100+ legacy ETL workflows (SSIS, PL/SQL, Informatica) to **modular dbt models** running on **Azure Data Lake Gen2**, enabling testability and version control.
- Led the design of a **Databricks Lakehouse** for Mayo Clinic's enterprise clinical warehouse using **Delta Lake**, allowing versioned updates and ML reproducibility.
- Built hybrid pipelines for HCA's multi-region cloud migration, integrating **on-prem Epic/Cerner extracts**, **AWS S3 ingestion**, and **Azure Blob Storage** via **Kafka Connect**.
- Orchestrated batch and streaming jobs using **Azure Data Factory**, **Azure Functions**, and **Synapse Pipelines** to support real-time data delivery for clinical dashboards and ML scoring.
- Built clinical analytics marts in **Snowflake** for surgical outcomes, readmission trends, and quality metrics across 180+ HCA hospitals.

• Building Data Pipelines

- Built and maintained 100+ data pipelines using **Apache Spark (batch + streaming)**, **Airflow**, **Python**, and **dbt**, handling EHR events, HL7 messages, and real-time device telemetry.
- Engineered **Kafka-based ingestion** of patient monitoring data, enabling sub-10s latency for critical care alerts and early warning systems.
- Designed robust backfill logic using **Delta Lake MERGE** and **dbt snapshots** to support longitudinal patient analyses without breaking downstream reporting.
- Developed **cloud-native transformation layers in Snowflake**, integrating structured and semi-structured data across clinical, scheduling, and patient satisfaction domains.

• Supporting ML Teams & AI Workflows

- Collaborated with ML engineers to build feature pipelines using **dbt + MLflow**, supporting models for **sepsis risk prediction**, **bedside deterioration**, and **no-show forecasting**.
- Enabled real-time model scoring with **Structured Streaming** and **MLflow Model Registry**, pushing predictions into clinical decision support systems.

- Built cohort definition logic for ML training datasets using **ICD-10 codes, LOINC, CPT**, and HL7 events to ensure temporal integrity and clinical relevance.
- Developed **OCR** pipelines to extract data from scanned intake forms and faxed referrals using **OpenCV, Tesseract, and spaCy**, supporting structured downstream ingestion.
- Delivered **PHI-masked datasets** to fine-tune **LLMs (e.g., GPT-3, BioBERT)** for summarization of discharge notes and patient instructions.
- Designed **A/B testing pipelines** to evaluate the impact of ML-driven triage and outreach workflows.
- **Security, Compliance & Governance**
 - Applied field-level encryption, dynamic masking, and **role-based access control** across Spark, Databricks, and Snowflake to meet **HIPAA, GxP, and SOC2 Type II** requirements.
 - Integrated **Great Expectations** for validation and embedded them into **CI/CD pipelines via GitHub Actions**, enforcing quality gates at every deployment.
 - Created comprehensive data lineage documentation using **dbt Docs, OpenLineage**, and **Azure Purview**, supporting audit needs for CMS, HEDIS, and Joint Commission reporting.
- **Automation, Monitoring & Reliability**
 - Implemented **GitOps workflows** using **Terraform, Docker, and Kubernetes** to automate deployment of dbt and Airflow pipelines across environments.
 - Monitored job health via **Prometheus + Grafana**, with custom metrics and alerting from Airflow and Spark — reducing incident response time by 50%.
- **Teamwork, Leadership & RCA**
 - Mentored 6+ junior engineers across multiple time zones, focusing on **dbt modularity, Spark performance tuning**, and **clinical data modeling**.
 - Acted as SME on **platform modernization for Mayo Clinic** and **ML governance for HCA Healthcare**, working closely with architects, clinical informaticists, and ML leads.
 - Led root cause analysis for production incidents, introducing **retry, rollback, and alerting** strategies that prevented repeat failures.

Comcast, Senior Data Analyst

Dec 2019 – Feb 2022 | Philadelphia, PA

Contributed to the **Xfinity Data & Insights team** by leading efforts to modernize analytics workflows, enhance customer experience visibility, and support predictive modeling across Xfinity's broadband, mobile, and home service lines. Directed initiatives to centralize product, customer, and support data within cloud-based platforms, enabling enterprise-grade insights, scalable data governance, and machine learning readiness throughout the Xfinity ecosystem.

- **Cloud Analytics Modernization & ETL Automation**
 - Migrated legacy SSIS and PL/SQL ETL jobs powering Xfinity reporting to **Azure Data Factory (ADF)** and **dbt**, reducing manual failures and enabling Git-based change tracking.
 - Built reusable **dbt models** over **Azure SQL** and **PostgreSQL**, capturing key metrics like device uptime, plan utilization, support ticket trends, and churn risk factors.
 - Designed ADF pipelines to ingest multi-source data — including **call center logs, device telemetry**, and **billing events** — into **Azure Data Lake Gen2** for analytics and monitoring.
 - Modeled key **fact/dimension tables** for customer activity, billing cycles, and support tickets using **star schema**, optimized for Power BI dashboards.
 - Integrated **Snowflake** as the primary data warehouse for unified analytics across product usage, customer support, and NPS tracking.
- **Analytics Enablement & Insights Engineering**
 - Created feature datasets to support **predictive models** on **churn, service degradation**, and **call volume forecasting**, partnering with data science teams.
 - Applied **Python, dbt, and T-SQL** to engineer time-based cohorts and event-driven labels tied to account activity, outages, and device error codes.
 - Automated profiling scripts to monitor **schema drift, missing values**, and **data anomalies** across 50+ pipelines, ensuring clean data for insights and modeling.
 - Enabled self-service reporting by delivering **Power BI-ready datasets** on product health, customer journeys, and agent performance metrics.
 - Supported integration with **Snowflake Cortex** to prepare time-series and anomaly detection-ready datasets used in ML pipelines.
- **Governance, Data Quality & Compliance**

- Developed **custom dbt tests** for data quality (nulls, types, values) and embedded them into **CI/CD pipelines** via Azure DevOps.
- Documented lineage, definitions, and update frequency in **dbt Docs**, enabling discoverability for 15+ teams across data science, BI, and product operations.
- Implemented **row- and field-level access controls** in **Snowflake and ADF** to protect PII while enabling role-based analytics across teams.
- Helped QA and validate new Xfinity data models during platform migrations and stakeholder adoption rollouts.
- **Collaboration, Operations & Stakeholder Support**
 - Acted as data liaison between **engineering, analytics, and product teams** to ensure KPI alignment across Xfinity broadband and mobile lines.
 - Built nightly **Python CLI tools** to validate schema consistency, detect distribution skews, and flag anomalies in raw vs. curated layers.
 - Supported BI developers by delivering **curated views** for dashboards monitoring **installation trends, tech support response times, and billing issue resolutions**.

CareFirst, Data Engineer

Sep 2015 – Nov 2019 | Baltimore, MD

Built scalable data pipelines and enabled real-time analytics for member behavior, claims processing, and care coordination initiatives. Supported machine learning and reporting workflows across CareFirst's cloud data infrastructure, helping transform healthcare operations and improve patient experience through data-driven insights.

- **Data Pipelines & ETL Development**
 - Developed **PySpark-based ETL pipelines** to ingest and transform large volumes of claims, eligibility, provider interactions, and call center data, supporting risk analytics and member engagement models.
 - Maintained and scheduled **Apache Airflow** batch jobs to deliver curated datasets for downstream clinical dashboards, care gap alerts, and utilization reviews.
 - Created modular **Python** utilities for schema validation, data flattening, and error handling, standardizing pipeline architecture across ingest jobs.
 - Optimized **Spark** jobs using memory tuning, broadcast joins, and partitioning strategies, improving processing performance by 40%+ on high-volume healthcare data.
- **Cloud Migration & Infrastructure Engineering**
 - Led migration of legacy **Hive/Hadoop workloads** to **Google Cloud Platform (GCP)** – including **BigQuery, Cloud Storage, and DataProc** – enabling faster development and lower compute costs.
 - Collaborated with infrastructure teams to define data lifecycle policies, access controls, and **HIPAA-compliant** storage retention on **GCS** and **BigQuery** datasets.
 - Set up **Airflow**-based observability using custom alerts, Slack integrations, and **Prometheus-style** monitoring, reducing incident resolution time by over 50%.
- **Real-Time Data & ML Pipeline Support**
 - Built real-time ingestion pipelines using **Kafka, Spark Streaming, and Google Pub/Sub** to capture claims intake events, care coordination updates, and member portal usage with low latency.
 - Partnered with data scientists to prepare training datasets for readmission prediction, high-risk member scoring, and service utilization forecasting, including feature engineering and cohort sampling.
 - Delivered pre-aggregated feature tables combining **EHR event data** and **support center logs**, accelerating model development cycles for the data science team.
- **Analytics Enablement & Cross-Team Delivery**
 - Created **KPI datasets** in **BigQuery** for tracking member engagement, chronic care management adherence, and digital tool adoption (e.g., telehealth usage).
 - Built **SQL**-based views and real-time dashboards to support clinical operations, provider outreach, and quality improvement teams.

CAI, Database Developer

Jul 2013 – Aug 2015 | Chicago, IL

Supported enterprise database development for various **City of Chicago** departments, including public health, licensing, and 311 services. Designed and optimized **SQL Server** databases and **ETL** workflows to support operational reporting, data integrity, and system modernization initiatives across city services.

- **Database Design & Development**
 - Designed and maintained normalized relational database schemas in **SQL Server 2012**, supporting applications for business licensing, inspections, and permit processing.

- Wrote complex **T-SQL stored procedures, functions, triggers, and views** to implement business logic and data validation rules across multiple departmental systems.
- Developed parameterized **SSRS reports** for internal and public-facing dashboards, including inspection compliance, license renewals, and case tracking.
- Refactored legacy database code and optimized query performance, reducing report generation time from minutes to seconds for key operational metrics.
- **ETL Development & Data Integration**
 - Built and maintained **ETL workflows** using **SQL Server Integration Services (SSIS)** to consolidate data from on-prem systems, Excel-based logs, and legacy Oracle databases into centralized SQL Server environments.
 - Created automated data load and transformation processes to feed **compliance** and **audit reporting pipelines** for departments such as Public Health and Buildings.
 - Implemented data quality checks, null handling, and deduplication logic within **ETL** pipelines to ensure reporting accuracy and regulatory compliance.
- **Support, Collaboration & Process Improvement**
 - Worked closely with city analysts and department heads to translate reporting needs into **SQL-based solutions**, including dashboards for code enforcement and service requests.
 - Participated in weekly sprint cycles with the **City's IT PMO** to support database changes for in-flight application development projects.
 - Supported database versioning and deployment automation using **Visual Studio Database Projects (SSDT)** and **TFS** for source control.

Languages & Programming

Python, SQL, Scala, Bash, Java, R

Data Warehousing & Lakehouse

Databricks, Delta Lake, Apache Iceberg, Apache Hudi, Snowflake, BigQuery, Redshift, Azure Synapse Analytics, Dremio, Trino, Starburst, Presto, HDFS

Machine Learning, AI & MLOps

MLflow, OpenLineage, Feast, Tecton, Metaflow, Kubeflow, SageMaker, Vertex AI, Azure ML, TensorFlow Extended (TFX), ONNX, Hugging Face Transformers, LangChain, LlamaIndex, BentoML, FastAPI, TorchServe, KServe, Airflow ML Decorators

BI, Semantic Layers & Analytics

Looker, LookML, Power BI, Tableau, Superset, Metabase, Cube.dev, dbt Semantic Layer, ThoughtSpot, Hex

Security, Compliance & Data Governance

HIPAA, SOC2, GDPR, PII Masking, Field-level Encryption, Audit Logging, Role-Based Access Control (RBAC), Unity Catalog, Immuta, Alation, Collibra, Great Expectations, Soda, Deequ, Monte Carlo, Datafold

DevOps, CI/CD & Infrastructure

Docker, Kubernetes, Terraform, Helm, Git, GitHub Actions, Jenkins, CircleCI, Argo CD, Vault

Execution & Strategic Thinking

- Agile & Scrum participation
- RCA and incident response ownership
- Strategic thinking and platform visioning
- Adaptability in fast-paced, cloud-native environments

Data Engineering & Orchestration

Apache Spark, Spark Structured Streaming, Apache Kafka, Kafka Connect, Apache Flink, Apache NiFi, Apache Beam, Google Cloud Dataflow, Apache Airflow, Cloud Composer, Azure Data Factory (ADF), AWS Glue, dbt, dbt Cloud, Talend, Informatica, Fivetran, Stitch, Airbyte, Meltano, Singer, Logstash, Prefect, Dagster, Python/Scala ingestion scripts

Cloud Platforms & Storage

AWS, Azure, GCP, AWS S3, Azure Data Lake Storage (ADLS), GCP Cloud Storage, AWS RDS, Azure SQL, Google Cloud Bigtable, Amazon DynamoDB, Azure Cosmos DB, Firestore

Feature Stores & Model Serving

Feast, Tecton, Vertex AI Feature Store, SageMaker Feature Store, BentoML, TorchServe, KServe, FastAPI, Flask

Data Ingestion & ETL/ELT Tools

SSIS, Talend, Informatica, Fivetran, Stitch, dbt, Apache NiFi, Logstash, Kafka Connect, custom Python/Scala ingestion scripts

Access Control, Monitoring & Observability

Prometheus, Grafana, Airflow Logs, CloudWatch, DataDog, Monte Carlo, Great Expectations, Soda, Metaplane, OpenTelemetry

Collaboration & Leadership

- Cross-functional collaboration with PMs, ML engineers, and business teams
- Stakeholder communication and technical storytelling
- Team leadership and mentoring
- Coaching junior engineers on best practices in Spark, dbt, Airflow, and ML pipelines
- Documentation and knowledge sharing